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(54) DEFORMABLE CONTAINER AND GAS STORAGE DEVICE

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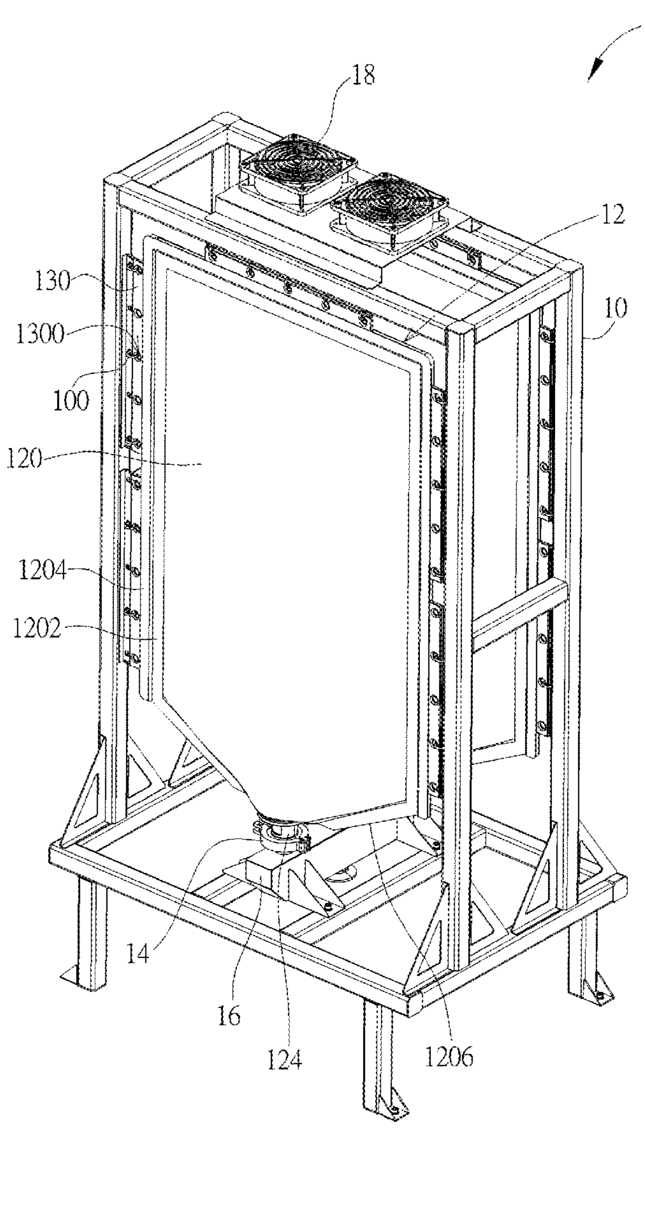
(57) **ABSTRACT**

A deformable container includes a bag body, a first connecting member and a second connecting member. The bag body is formed with an opening. The first connecting member has a first tube portion and a first annular flange protruding from an end of the first tube portion. The first tube portion passes through the opening to be exposed from the bag body. The bag body is pressed against a pressing surface of the first annular flange. The second connecting member has a second tube portion. The second tube portion is sleeved on the first tube portion.



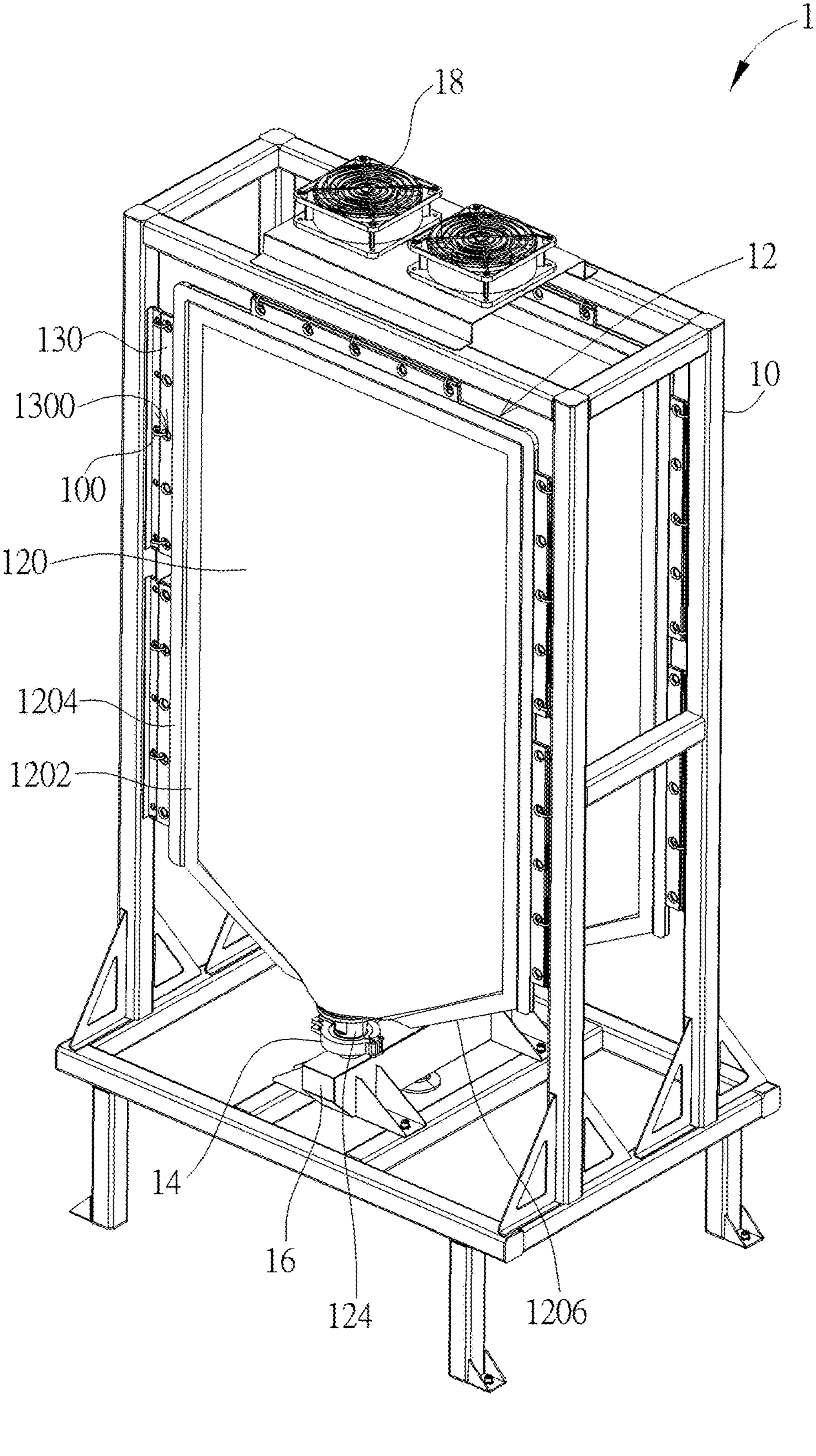


FIG. 1

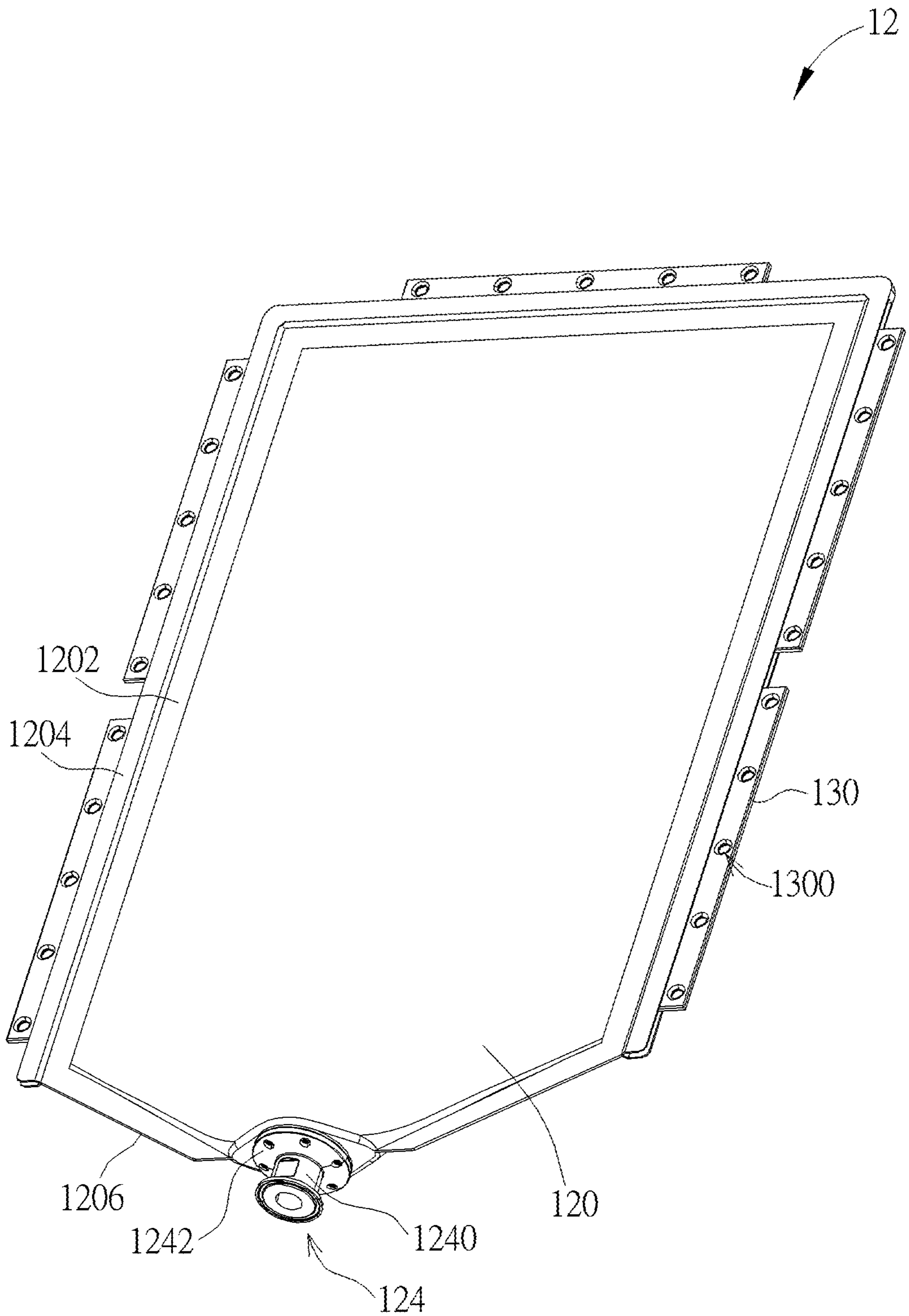


FIG. 2

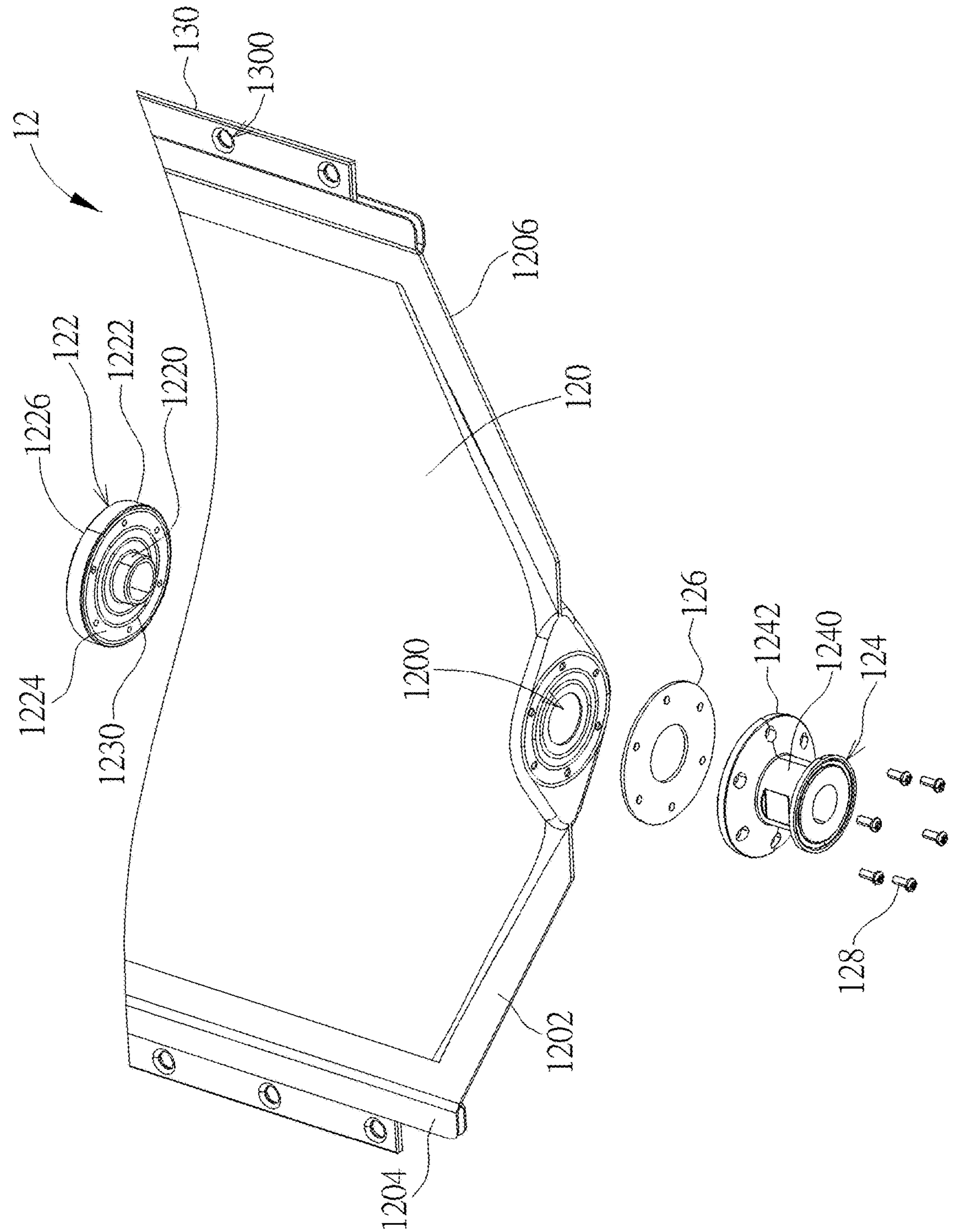

12
130
1300
1206
120
122
1222
1220
1226
1224
1230
1200
126
1242
1240
124
128
1202
1204

FIG. 3

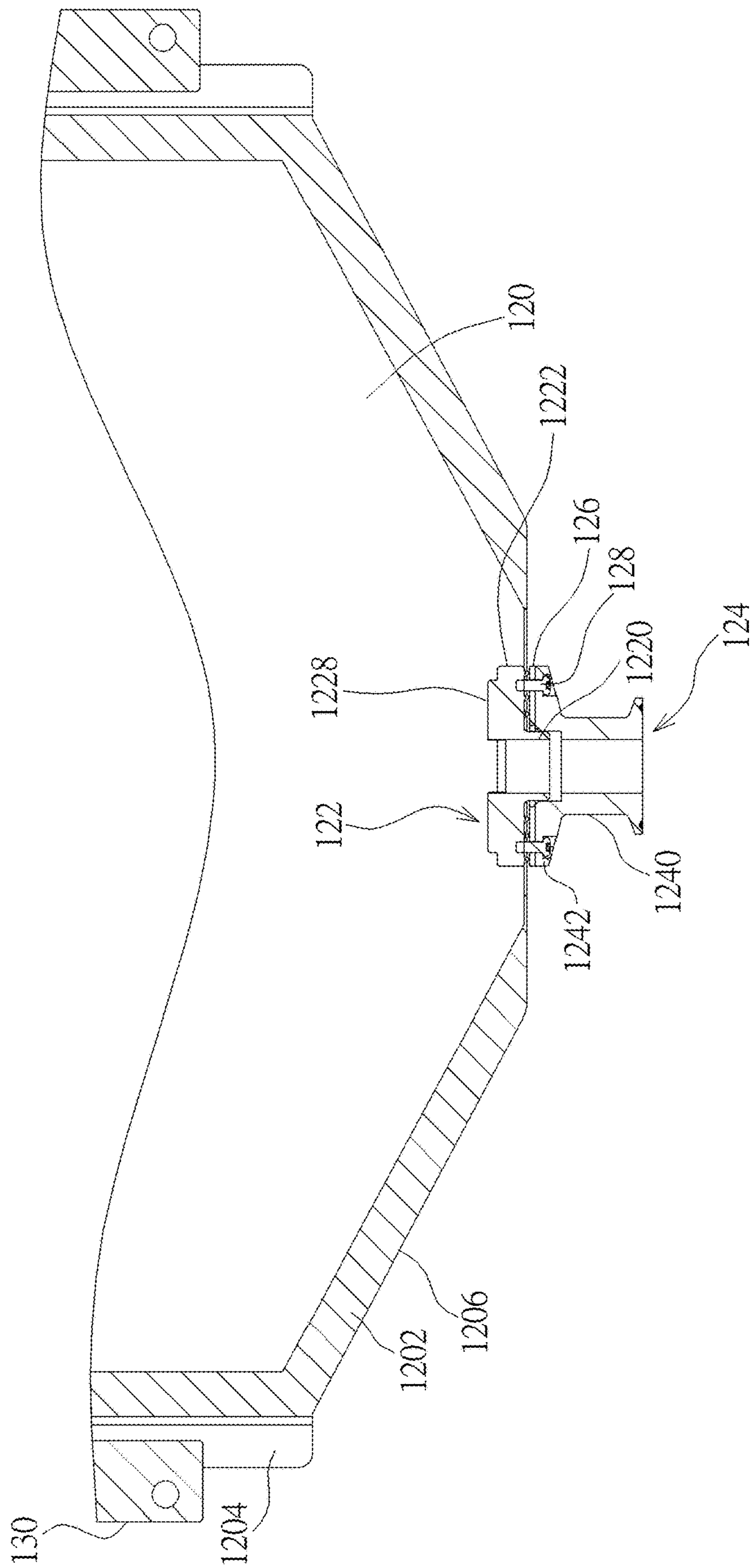

12
120
1222
126
128
1220
124
1228
122
1240
1242
1206
1202
130
1204

FIG. 4

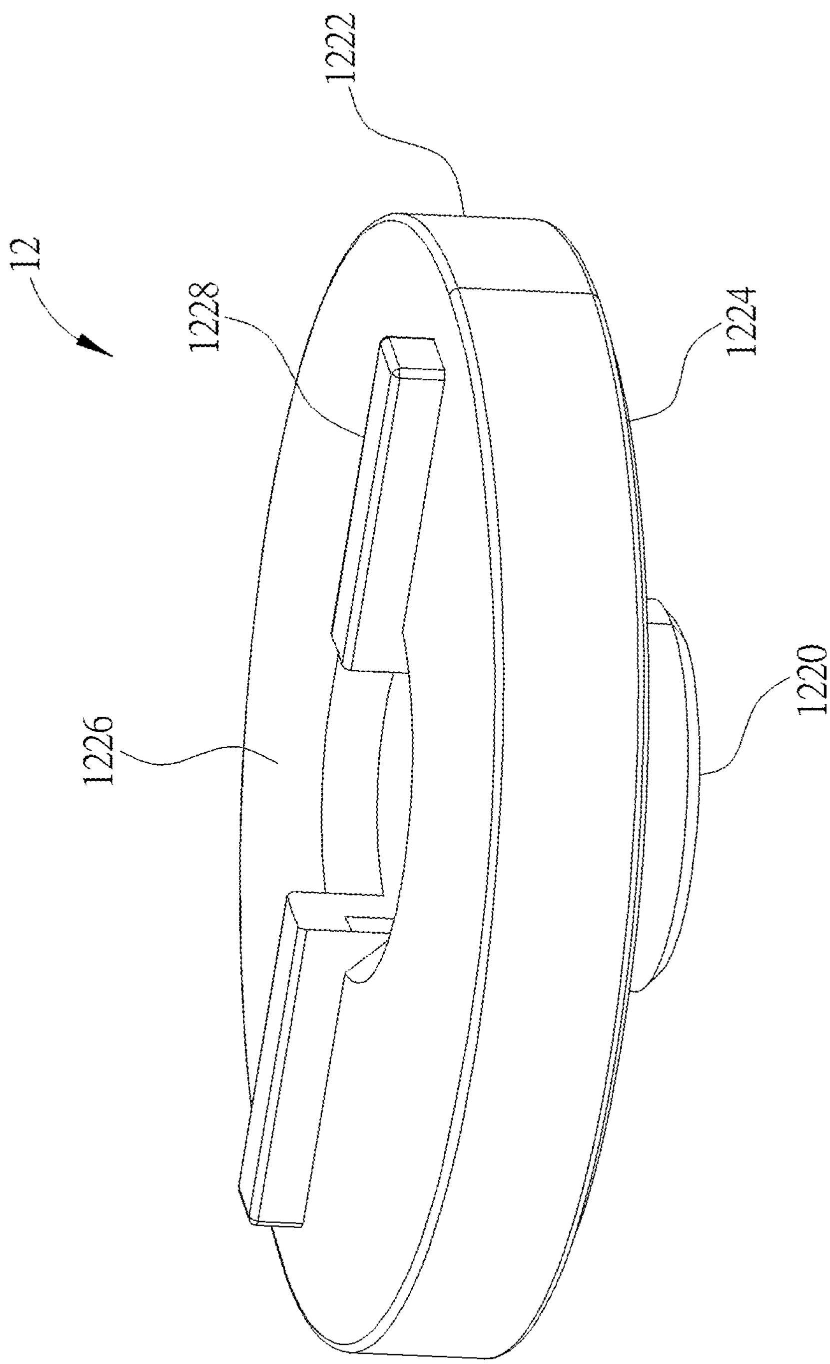


FIG. 5

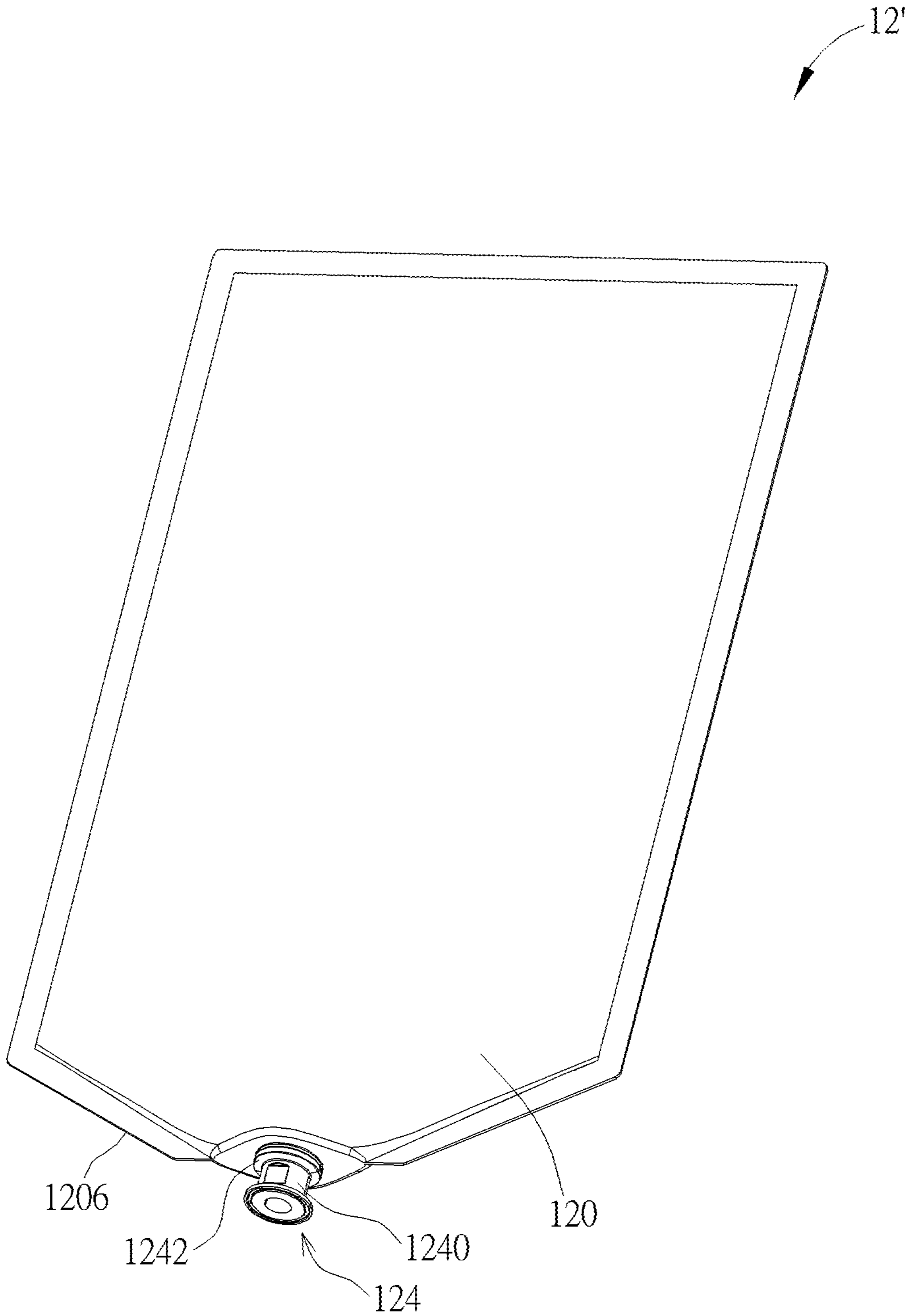


FIG. 6

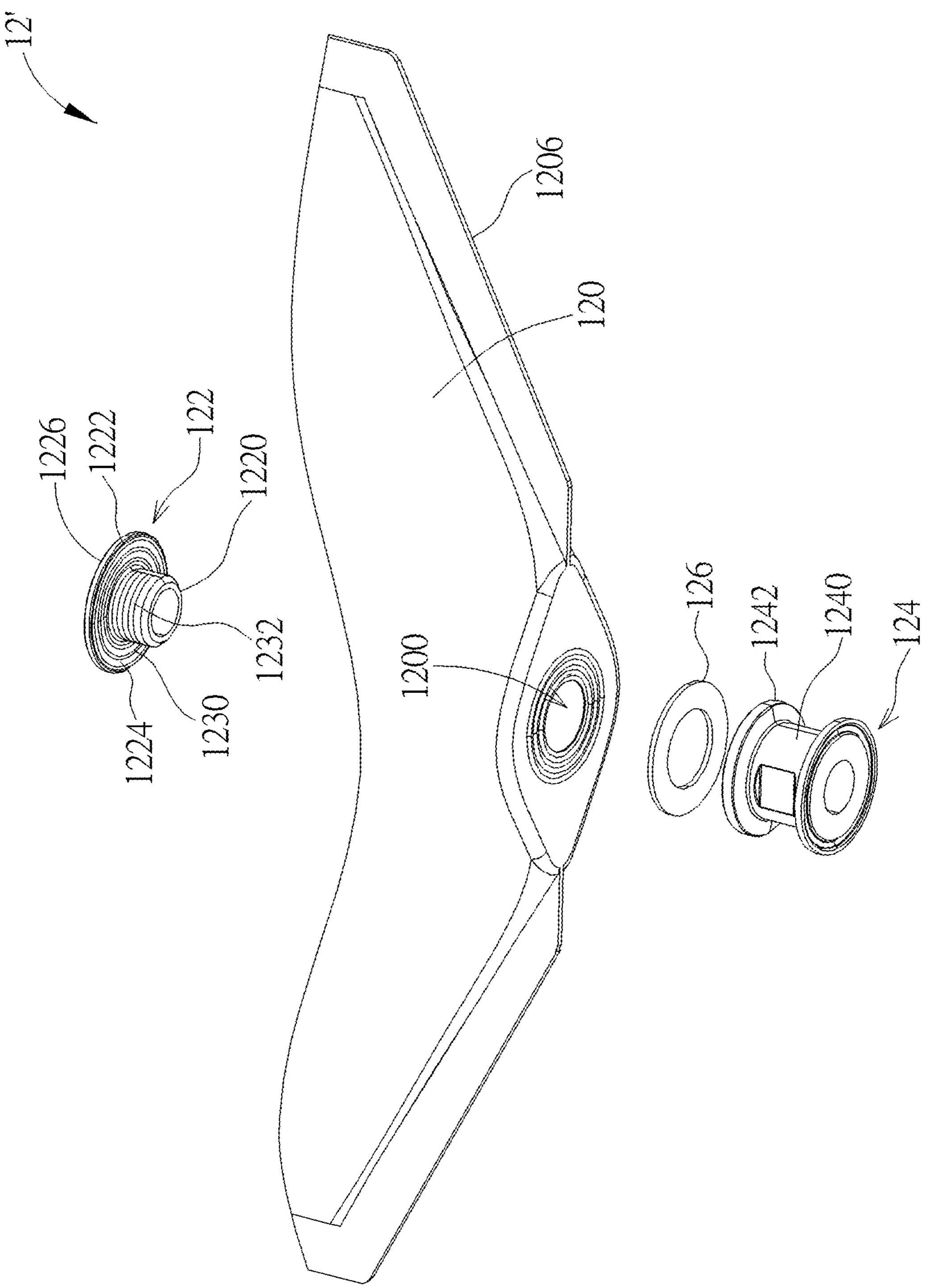

12'
1206
120
122
1226
1222
1220
1232
1230
1224
1200
126
1242
1240
124

FIG. 7

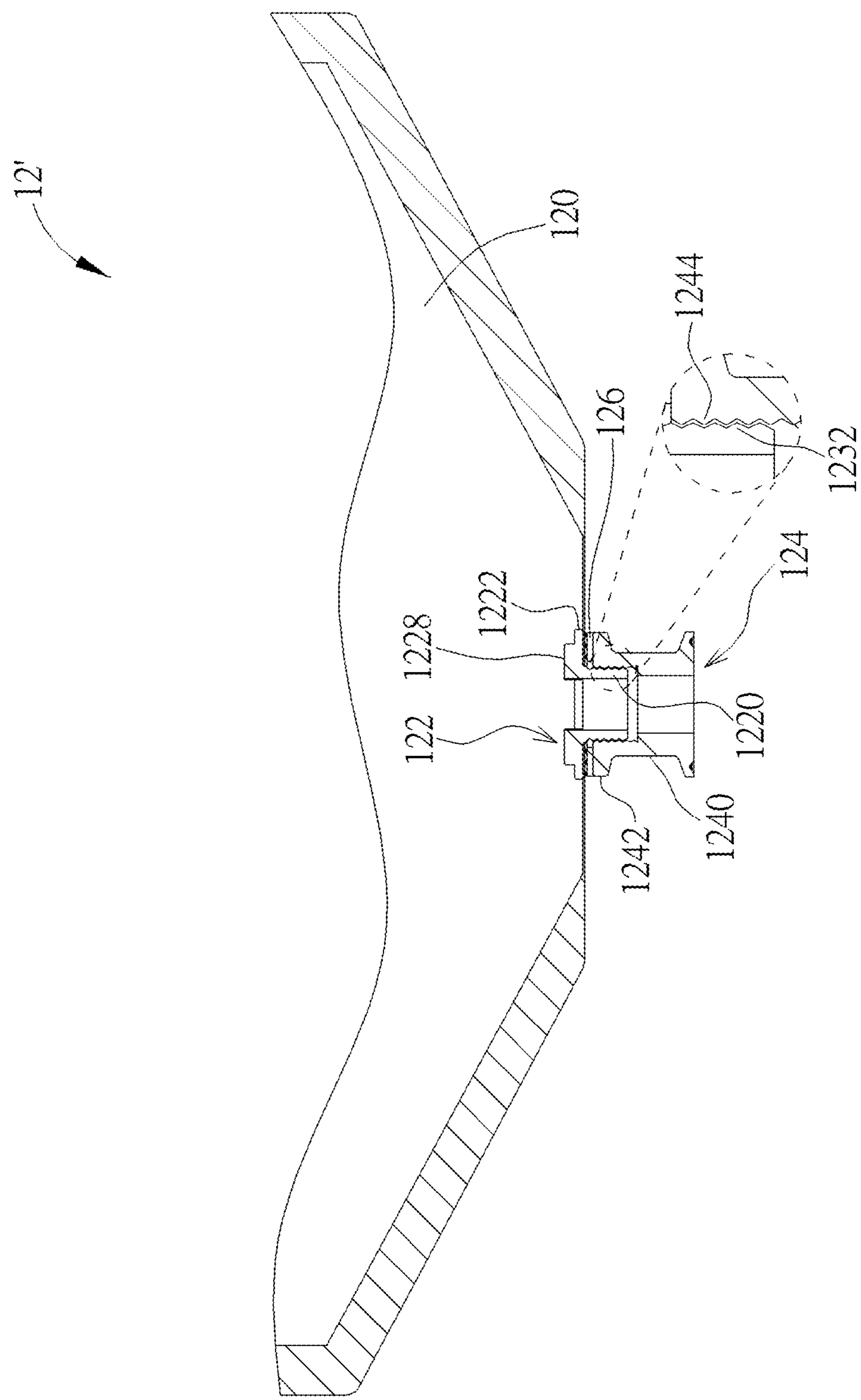


FIG. 8

DEFORMABLE CONTAINER AND GAS STORAGE DEVICE

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0001] The invention relates to a deformable container and a gas storage device and, more particularly, to a deformable container capable of effectively preventing gas from leaking and a gas storage device equipped with the deformable container.

2. Description of the Prior Art

[0002] An immersion cooling system uses a cooling liquid to dissipate heat from electronic components by a phase change manner. The cooling liquid evaporates into gas after absorbing the heat generated by the electronic component. Thus, the immersion cooling system is usually equipped with a gas storage device for temporarily storing the gas of the cooling liquid and controlling the pressure. A container of the gas storage device is connected to a cooling tank by a tube. However, the joint surface between the container of the gas storage device and the tube is easily broken due to expansion and contraction, thereby resulting in gas leakage.

SUMMARY OF THE INVENTION

[0003] According to an embodiment of the invention, a deformable container comprises a bag body, a first connecting member and a second connecting member. The bag body is formed with an opening. The first connecting member has a first tube portion and a first annular flange protruding from an end of the first tube portion. The first tube portion passes through the opening to be exposed from the bag body. The bag body is pressed against a pressing surface of the first annular flange. The second connecting member has a second tube portion. The second tube portion is sleeved on the first tube portion.

[0004] According to another embodiment of the invention, a gas storage device comprises a frame and a deformable container. The deformable container is connected to the frame. The deformable container comprises a bag body, a first connecting member and a second connecting member. The bag body is formed with an opening. The first connecting member has a first tube portion and a first annular flange protruding from an end of the first tube portion. The first tube portion passes through the opening to be exposed from the bag body. The bag body is pressed against a pressing surface of the first annular flange. The second connecting member has a second tube portion. The second tube portion is sleeved on the first tube portion.

[0005] These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. **1** is a perspective view illustrating a gas storage device according to an embodiment of the invention.

[0007] FIG. **2** is a perspective view illustrating a deformable container shown in FIG. **1**.

[0008] FIG. **3** is a partial exploded view illustrating the deformable container shown in FIG. **2**.

[0009] FIG. **4** is a partial sectional view illustrating the deformable container shown in FIG. **2**.

[0010] FIG. **5** is a perspective view illustrating a first connecting member shown in FIG. **3** from another viewing angle.

[0011] FIG. **6** is a perspective view illustrating a deformable container according to another embodiment of the invention.

[0012] FIG. **7** is a partial exploded view illustrating the deformable container shown in FIG. **6**.

[0013] FIG. **8** is a partial sectional view illustrating the deformable container shown in FIG. **6**.

DETAILED DESCRIPTION

[0014] Referring to FIGS. **1** to **5**, FIG. **1** is a perspective view illustrating a gas storage device **1** according to an embodiment of the invention, FIG. **2** is a perspective view illustrating a deformable container **12** shown in FIG. **1**, FIG. **3** is a partial exploded view illustrating the deformable container **12** shown in FIG. **2**, FIG. **4** is a partial sectional view illustrating the deformable container **12** shown in FIG. **2**, and FIG. **5** is a perspective view illustrating a first connecting member **122** shown in FIG. **3** from another viewing angle.

[0015] As shown in FIG. **1**, the gas storage device **1** comprises a frame **10** and a deformable container **12**. The deformable container **12** is connected to the frame **10**. The gas storage device **1** uses the frame **10** to support the deformable container **12**, which may keep the deformable container **12** upright to prevent the deformable container **12** from tipping over and affecting its expansion and contraction performance. It should be noted that the structure of the frame **10** may be designed according to practical applications, so the invention is not limited to the embodiment shown in the figure.

[0016] In this embodiment, the deformable container **12** may be connected to an airtight base **16** through a quick release connector **14**, and the airtight base **16** may be connected to a cooling tank of an immersion cooling system (not shown) through a tube (not shown). In general, the cooling tank may store a cooling liquid (e.g. dielectric liquid) with a low boiling point. An electronic component (not shown) may be immersed in the cooling liquid. The cooling liquid evaporates and changes into gas after absorbing the heat generated by the electronic component. The deformable container **12** is used to temporarily store the gas of the cooling liquid and control the pressure.

[0017] In this embodiment, two deformable containers **12** may be connected to the frame **10** and the airtight base **16** side by side. Furthermore, at least one fan **18** may be disposed on the top of the frame **10**, wherein the fan **18** may be disposed between the two deformable containers **12**. Accordingly, the downward airflow generated by the fan **18** may dissipate heat from the two deformable containers **12** at the same time. It should be noted that the number of frames **10**, deformable containers **12**, quick release connectors **14**, airtight bases **16** and fans **18** may be adjusted according to practical applications, which make the gas storage device **1** expandable.

[0018] As shown in FIGS. **2** to **4**, the deformable containers **12** comprises a bag body **120**, a first connecting member **122**, a second connecting member **124**, a sealing member **126**, a plurality of fixing members **128** and a plurality of support members **130**. The bag body **120** is formed with an opening **1200**. In this embodiment, the bag body **120** may be made of plastic film and aluminum foil, the first connecting member **122** and the support members **130** may be made of plastic, the second connecting member **124** may be made of metal or plastic, the sealing member **126** may be an O-ring, and the fixing members **128** may be screws, but the invention is not so limited. In an embodiment, when the bag body **120** is made of plastic film and aluminum foil, the aluminum foil is located between the upper and lower layers of the plastic film, wherein the upper and lower layers of the plastic films are used to be connected with each other and the aluminum foil is used to block light. It should be noted that the aluminum foil may be omitted according to practical applications. In an embodiment, when both the second connecting member **124** and the quick release connector **14** are made of metal, the connecting strength between the second connecting member **124** and the quick release connector **14** may be enhanced.

[0019] The first connecting member **122** has a first tube portion **1220** and a first annular flange **1222** protruding from an end of the first tube portion **1220**. The first tube portion **1220** of the first connecting member **122** passes through the opening **1200** of the bag body **120** to be exposed from the bag body **120**. Furthermore, the bag body **120** is pressed against a pressing surface **1224** of the first annular flange **1222** of the first connecting member **122**.

[0020] In this embodiment, a non-pressing surface **1226** of the first annular flange **1222** may have a positioning structure **1228**, wherein the non-pressing surface **1226** is opposite to the pressing surface **1224**. In the invention, the positioning structure **1228** of the first connecting member **122** may be inserted into a positioning hole of a fixture (not shown) first. Then, before the periphery of the bag body **120** is pressed, the opening **1200** of the bag body **120** is sleeved on the first tube portion **1220** of the first connecting member **122**. Then, the bag body **120** and the pressing surface **1224** of the first annular flange **1222** are hot pressed and then the periphery of the bag body **120** is hot pressed. Accordingly, the first tube portion **1220** of the first connecting member **122** passes through the opening **1200** of the bag body **120** to be exposed from the bag body **120**, and the first annular flange **1222** of the first connecting member **122** is wrapped in the bag body **120**. After the bag body **120** and the pressing surface **1224** of the first annular flange **1222** of the first connecting member **122** are hot pressed, the bag body **120** will be tightly connected with the pressing surface **1224** of the first annular flange **1222**, thereby enhancing the sealing effect between the bag body **120** and the first connecting member **122** and effectively preventing gas leakage. Furthermore, since the bag body **120** and the pressing surface **1224** of the first annular flange **1222** are pressed together in a surface contact manner, the connecting strength between the bag body **120** and the first annular flange **1222** of the first connecting member **122** will be effectively enhanced, so as to prevent the joint between the bag body **120** and the first annular flange **1222** of the first connecting member **122** from breaking due to expansion and contraction.

[0021] In this embodiment, the pressing surface **1224** of the first annular flange **1222** may have a protruding structure **1230**, as shown in FIG. **3**. The bag body **120** may be pressed against the protruding structure **1230** to increase the contact area between the bag body **120** and the first annular flange **1222**, so as to enhance the connecting strength between the bag body **120** and the first annular flange **1222**. In this embodiment, the protruding structure **1230** may be composed of a plurality of protruding ribs arranged in concentric circles, but the invention is not so limited.

[0022] The second connecting member **124** has a second tube portion **1240** and a second annular flange **1242** protruding from an end of the second tube portion **1240**. After the bag body **120** and the first annular flange **1222** of the first connecting member **122** are hot pressed, the sealing member **126** is sleeved on the first tube portion **1220** of the first connecting member **122** and abuts against the bag body **120**. Then, the second tube portion **1240** of the second connecting member **124** is sleeved on the first tube portion **1220** of the first connecting member **122**, such that the sealing member **126** and the bag body **120** are sandwiched between the first annular flange **1222** of the first connecting member **122** and the second annular flange **1242** of the second connecting member **124**. Then, the fixing members **128** pass through the second annular flange **1242**, the sealing member **126** and the bag body **120** in sequence to be fixed to the first annular flange **1222**, such that the second annular flange **1242** presses the sealing member **126** and the bag body **120** tightly. Then, the second connecting member **124** may be connected to the airtight base **16** through the quick release connector **14**, as shown in FIG. **1**. The sealing member **126** can prevent gas from leaking from a gap between the first connecting member **122** and the second connecting member **124**.

[0023] In this embodiment, the first annular flange **1222** may be formed with a plurality of fixing holes (e.g. screw holes), and the second annular flange **1242**, the sealing member **126** and the bag body **120** may be formed with a plurality of through holes corresponding to the fixing holes, such that the fixing members **128** may pass through the through holes of the second annular flange **1242**, the sealing member **126** and the bag body **120** in sequence to be fixed to the fixing holes of the first annular flange **1222**.

[0024] The support members **130** are pressed against a periphery of the bag body **120**. In this embodiment, the periphery of the bag body **120** may be formed with a first pressing region **1202** and a second pressing region **1204**, wherein the second pressing region **1204** surrounds the first pressing region **1202**. The support members **130** may be pressed against the second pressing region **1204**. In an embodiment, when hot pressing the periphery of the bag body **120**, a specific width (e.g. 30 cm) may be reserved around the bag body **120**. Then, the first pressing region **1202** is formed by hot pressing the upper and lower layers of the plastic film with a first width (e.g. 10 cm) of the specific width close to the inside of the bag body **120**. Then, a part of each support member **130** is placed between the upper and lower layers of the plastic film at a second width (e.g. 20 cm) of the specific width close to the outside of the bag body **120**. Then, the upper and lower layers of the plastic film are hot pressed to form the second pressing region **1204**, and another part of each support member **130** is exposed from the bag body **120**. Since the second pressing region **1204** surrounds the first pressing region **1202**, the first pressing region **1202** will not be broken due to the pulling of the support member **130**. Accordingly, the sealing effect of the first pressing region **1202** can be ensured.

[0025] As shown in FIG. **1**, each of the support members **130** may be formed with a plurality of hanging holes **1300**, and the frame **10** may comprise a plurality of hanging rings **100**. The hanging holes **1300** may be hung on the hanging rings **100**, such that the deformable container **12** is connected to the frame **10** by the support members **130**. Accordingly, the deformable container **12** may be hung on the frame **10** and keep upright to prevent the deformable container **12** from tipping over and affecting its expansion and contraction performance. It should be noted that the deformable container **12** may be connected to the frame **10** by other fixing or engaging structures, so the invention is not limited to the hanging manner.

[0026] In this embodiment, a bottom **1206** of the bag body **120** may be inclined with respect to the first tube portion **1220** of the first connecting member **122** to help recover the liquid condensed from the gas of the cooling liquid.

[0027] Referring to FIGS. **6** to **8**, FIG. **6** is a perspective view illustrating a deformable container **12'** according to another embodiment of the invention, FIG. **7** is a partial exploded view illustrating the deformable container **12'** shown in FIG. **6**, and FIG. **8** is a partial sectional view illustrating the deformable container **12'** shown in FIG. **6**.

[0028] In another embodiment, the aforesaid deformable container **12** may be replaced by the deformable container **12'** shown in FIGS. **6** to **8**. The main difference between the deformable container **12'** and the aforesaid deformable container **12** is that the first tube portion **1220** of the first connecting member **122** of the deformable container **12'** has an outer thread **1232**, and the second tube portion **1240** of the second connecting member **124** of the deformable container **12'** has an inner thread **1244**. The inner thread **1244** and the outer thread **1232** may be screwed with each other, such that the second connecting member **124** is fixed on the first connecting member **122** and the second annular flange **1242** presses the sealing member **126** and the bag body **120** tightly. Therefore, in this embodiment, the fixing members **128**, the through holes and the fixing holes mentioned above may be omitted. It should be noted that the deformable container **12'** may also be equipped with the aforesaid support members **130** to be connected with the frame **10** shown in FIG. **1**.

[0029] As mentioned above, since the bag body and the pressing surface of the first annular flange of the first connecting member are pressed together in a surface contact manner, the connecting strength between the bag body and the first annular flange of the first connecting member will be effectively enhanced, so as to prevent the joint between the bag body and the first annular flange of the first connecting member from breaking due to expansion and contraction. After the bag body and the pressing surface of the first annular flange of the first connecting member are pressed, the bag body will be tightly connected with the pressing surface of the first annular flange, thereby enhancing the sealing effect between the bag body and the first connecting member and effectively preventing gas leakage. Furthermore, the sealing member may be disposed between the first connecting member and the second connecting member to prevent gas from leaking from the gap between the first connecting member and the second connecting member. Moreover, the deformable container may be connected to the frame to keep upright, so as to prevent the deformable container **12** from tipping over and affecting its expansion and contraction performance.

[0030] Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

What is claimed is:

1. A deformable container comprising:
a bag body formed with an opening;
a first connecting member having a first tube portion and a first annular flange protruding from an end of the first tube portion, the first tube portion passing through the opening to be exposed from the bag body, the bag body being pressed against a pressing surface of the first annular flange; and
a second connecting member having a second tube portion, the second tube portion being sleeved on the first tube portion.

2. The deformable container of claim **1**, further comprising a sealing member sleeved on the first tube portion and abutting against the bag body, wherein the second connecting member further has a second annular flange protruding from an end of the second tube portion, and the sealing member and the bag body are sandwiched between the first annular flange and the second annular flange.

3. The deformable container of claim **2**, further comprising a plurality of fixing members, wherein the plurality of fixing members pass through the second annular flange, the sealing member and the bag body in sequence to be fixed to the first annular flange.

4. The deformable container of claim **1**, wherein the first tube portion has an outer thread, the second tube portion has an inner thread, and the inner thread and the outer thread are screwed with each other.

5. The deformable container of claim **1**, wherein the pressing surface has a protruding structure and the bag body is pressed against the protruding structure.

6. The deformable container of claim **1**, wherein a bottom of the bag body is inclined with respect to the first tube portion.

7. The deformable container of claim **1**, further comprising a plurality of support members pressed against a periphery of the bag body.

8. The deformable container of claim **7**, wherein the periphery of the bag body is formed with a first pressing region and a second pressing region, the second pressing region surrounds the first pressing region, and the plurality of support members are pressed against the second pressing region.

9. The deformable container of claim **7**, wherein each of the plurality of support members is formed with a plurality of hanging holes.

10. The deformable container of claim **1**, wherein a non-pressing surface of the first annular flange has a positioning structure and the non-pressing surface is opposite to the pressing surface.

11. A gas storage device comprising:
a frame; and
a deformable container connected to the frame, the deformable container comprising:
a bag body formed with an opening;
a first connecting member having a first tube portion and a first annular flange protruding from an end of the first tube portion, the first tube portion passing through the opening to be exposed from the bag body, the bag body being pressed against a pressing surface of the first annular flange; and

a second connecting member having a second tube portion, the second tube portion being sleeved on the first tube portion.

12. The gas storage device of claim **11**, wherein the deformable container further comprises a sealing member sleeved on the first tube portion and abutting against the bag body, the second connecting member further has a second annular flange protruding from an end of the second tube portion, and the sealing member and the bag body are sandwiched between the first annular flange and the second annular flange.

13. The gas storage device of claim **12**, wherein the deformable container further comprises a plurality of fixing members, and the plurality of fixing members pass through the second annular flange, the sealing member and the bag body in sequence to be fixed to the first annular flange.

14. The gas storage device of claim **11**, wherein the first tube portion has an outer thread, the second tube portion has an inner thread, and the inner thread and the outer thread are screwed with each other.

15. The gas storage device of claim **11**, wherein the pressing surface has a protruding structure and the bag body is pressed against the protruding structure.

16. The gas storage device of claim **11**, wherein a bottom of the bag body is inclined with respect to the first tube portion.

17. The gas storage device of claim **11**, wherein the deformable container further comprises a plurality of support members pressed against a periphery of the bag body, and the deformable container is connected to the frame by the plurality of support members.

18. The gas storage device of claim **17**, wherein the periphery of the bag body is formed with a first pressing region and a second pressing region, the second pressing region surrounds the first pressing region, and the plurality of support members are pressed against the second pressing region.

19. The gas storage device of claim **17**, wherein each of the plurality of support members is formed with a plurality of hanging holes, the frame comprises a plurality of hanging rings, and the plurality of hanging holes are hung on the plurality of hanging rings.

20. The gas storage device of claim **11**, wherein a non-pressing surface of the first annular flange has a positioning structure and the non-pressing surface is opposite to the pressing surface.

* * * * *